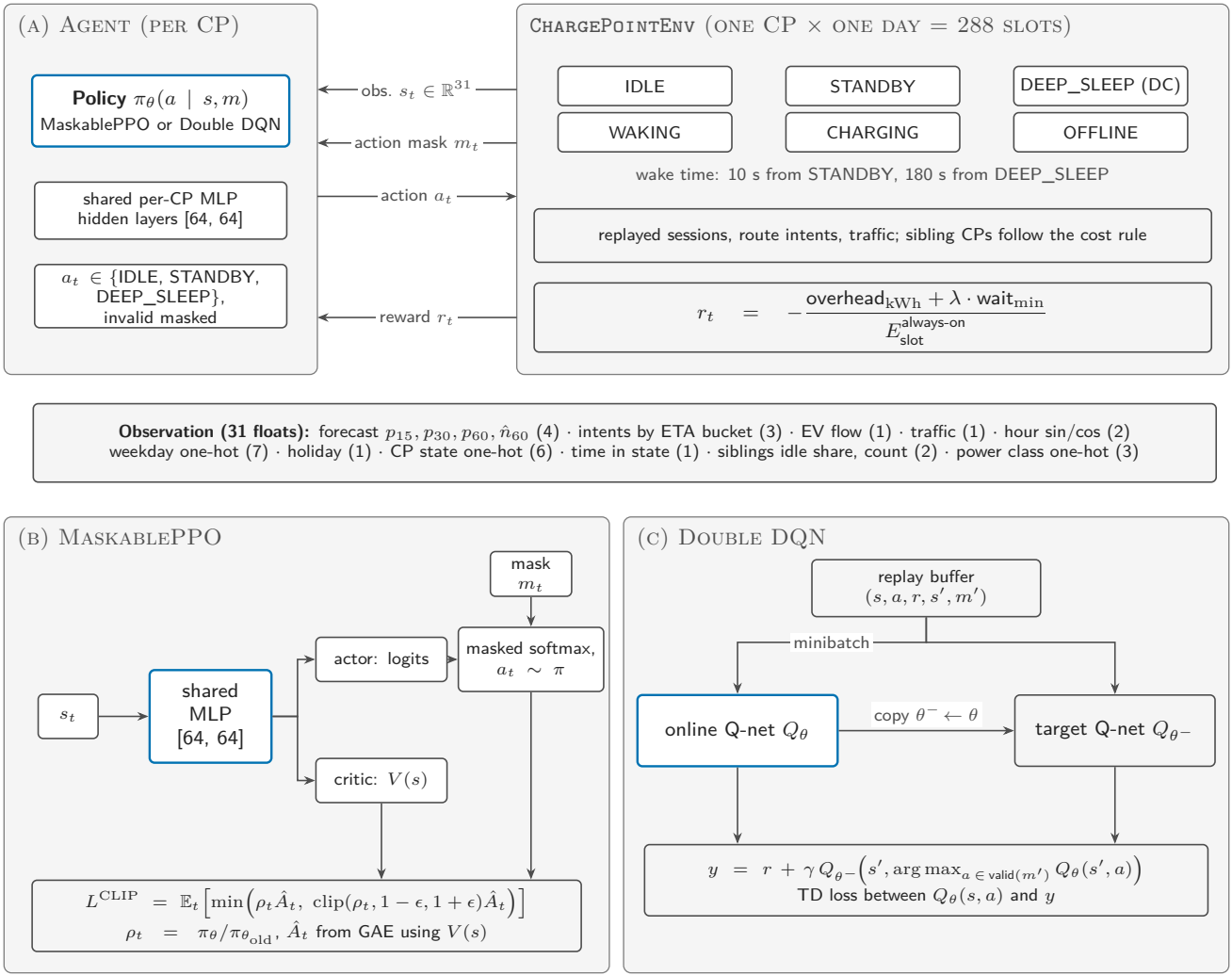

(A) Agent (per CP)
Policy $\pi_\theta(a \mid s, m)$
MaskablePPO or Double DQN
shared per-CP MLP
hidden layers [64, 64]
$a_t \in$ {IDLE, STANDBY, DEEP_SLEEP}, invalid masked
obs. $s_t \in \mathbb{R}^{31}$
action mask m_t
action a_t
reward r_t
ChargePointEnv (one CP × one day = 288 slots)
IDLE
STANDBY
DEEP_SLEEP (DC)
WAKING
CHARGING
OFFLINE
wake time: 10 s from STANDBY, 180 s from DEEP_SLEEP
replayed sessions, route intents, traffic; sibling CPs follow the cost rule
$r_t = -\frac{\text{overhead}_{\text{kWh}} + \lambda \cdot \text{wait}_{\text{min}}}{E_{\text{slot}}^{\text{always-on}}}$
Observation (31 floats): forecast $p_{15}, p_{30}, p_{60}, \hat{n}_{60}$ (4) · intents by ETA bucket (3) · EV flow (1) · traffic (1) · hour sin/cos (2) weekday one-hot (7) · holiday (1) · CP state one-hot (6) · time in state (1) · siblings idle share, count (2) · power class one-hot (3)
(B) MaskablePPO
mask m_t
s_t
shared MLP [64, 64]
actor: logits
masked softmax, $a_t \sim \pi$
critic: $V(s)$
$L^{\text{CLIP}} = \mathbb{E}_t\left[\min\left(\rho_t \hat{A}_t,\ \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right]$
$\rho_t = \pi_\theta / \pi_{\theta_{\text{old}}}$, $\hat{A}_t$ from GAE using $V(s)$
(C) Double DQN
replay buffer (s, a, r, s', m')
minibatch
online Q-net Q_θ
copy $\theta^- \leftarrow \theta$
target Q-net Q_{θ^-}
$y = r + \gamma\, Q_{\theta^-}\left(s', \arg\max_{a \in \text{valid}(m')} Q_\theta(s', a)\right)$
TD loss between $Q_\theta(s, a)$ and y